
Lecture 13: Support Vector Machine

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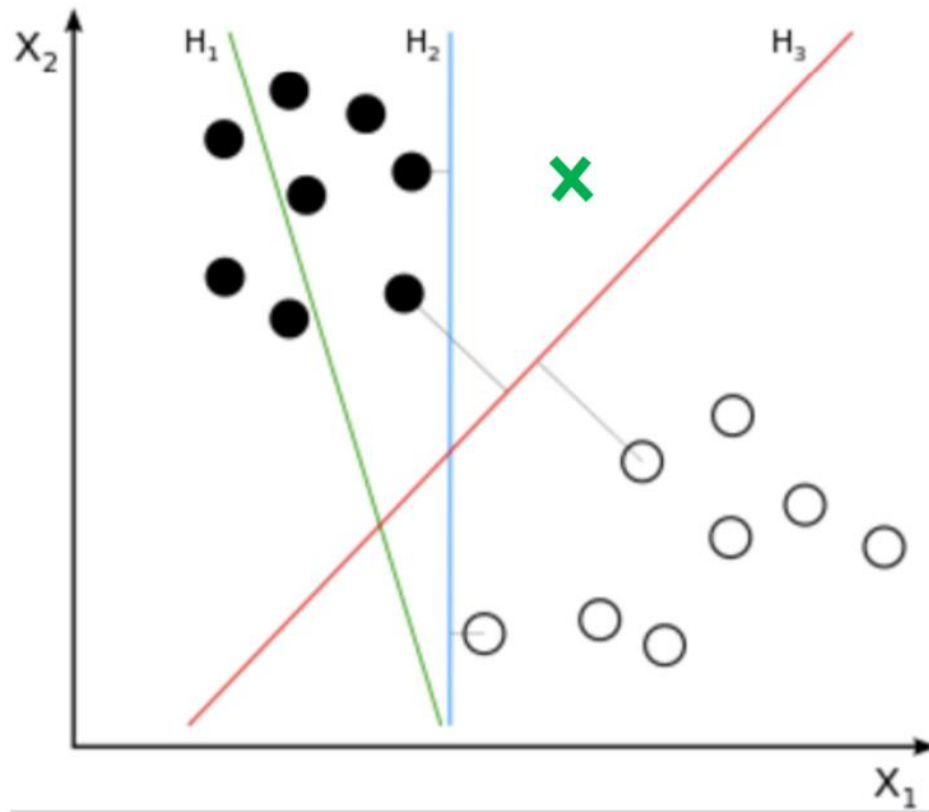


Support Vector Machine

- A Support Vector Machine (SVM) is a powerful and versatile Machine Learning model
- It is capable of performing **linear** or **nonlinear classification**, **regression**, and even **outlier detection**
- SVMs are particularly well suited for classification of complex small- or medium-sized datasets



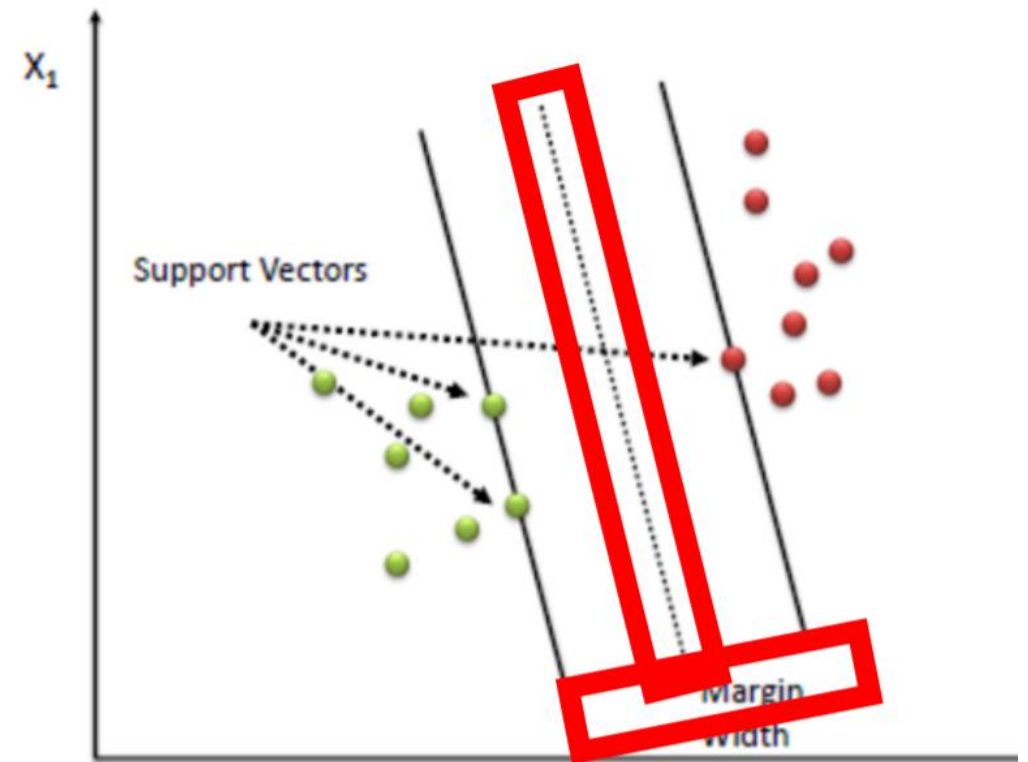
Decision Boundary For Linear Classification



- To which class would each decision boundary assign the new instance?
- Which decision boundary would you choose to separate data?

Hyperplane and Margin

- Choose **hyperplane** that maximizes the “margin” width
- Margin $\rightarrow$ distance between the decision boundary (i.e., parallel to hyperplane) and training instances (“support vectors”) closest to the hyperplane
- SVM: creates BOTH hyperplane AND margin
- Support vectors: controls/defines the margin



Support Vector Machine

- Linear or nonlinear classification, regression, outlier detection
- Excellent for small or medium sized datasets
- Large margin classification - fit the widest possible street between the classes
- More instances “off-the-street” does not affect decision boundary



Decision Boundary of SVM

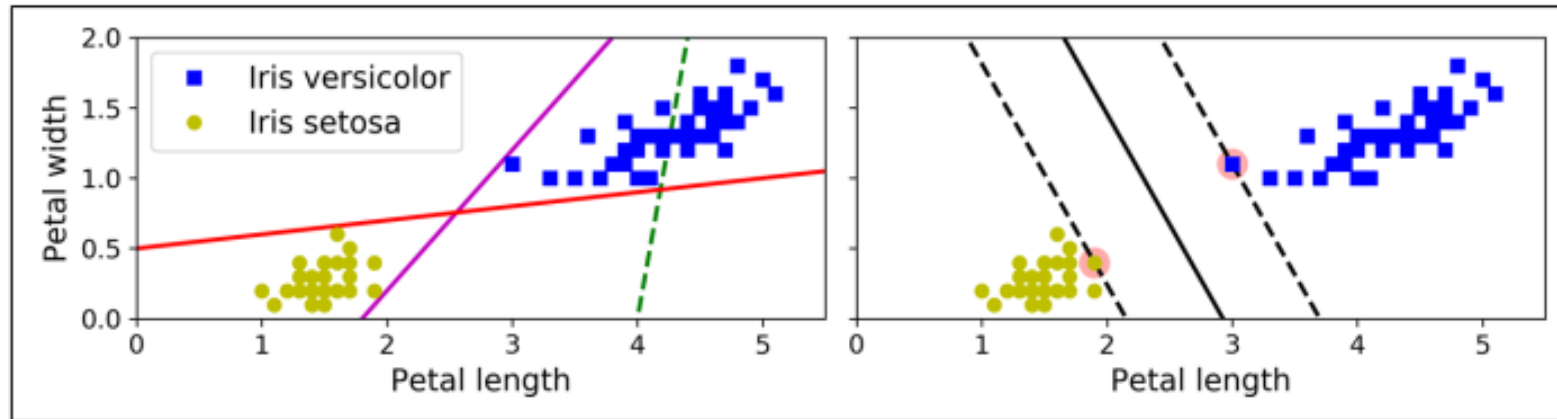


Figure 5-1. Large margin classification

- The left plot shows the decision boundaries of three possible linear classifiers. Dashed one is not useful
- The other two models work perfectly on this training set, but their decision boundaries come so close to the instances that these models will probably not perform as well on new instances.
- In contrast, the solid line in the plot on the right represents the decision boundary of an SVM classifier; this line **not only separates the two classes** but also stays as **far away from the closest training instances** as possible.
- This is called ***large margin classification***

Feature Scale

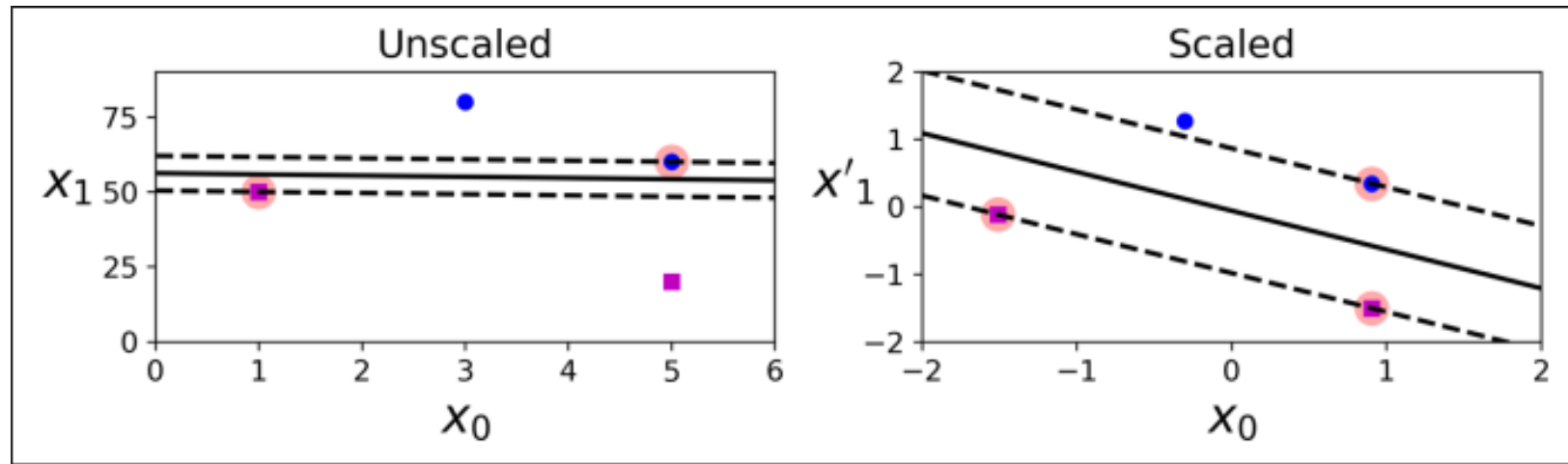
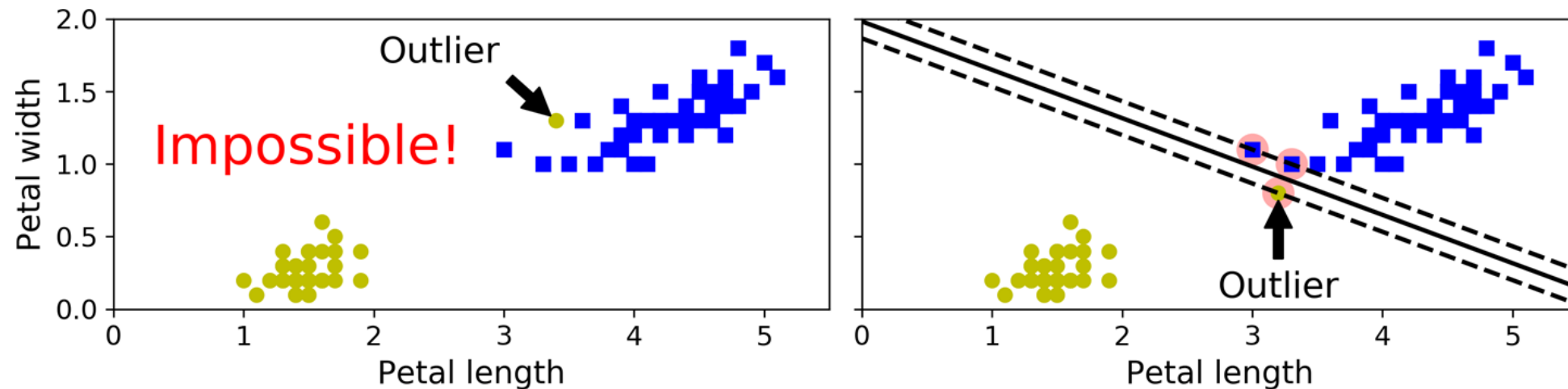


Figure 5-2. Sensitivity to feature scales

- SVMs are sensitive to the feature scales
- in the left plot, the vertical scale is much larger than the horizontal scale, so the widest possible street is close to horizontal
- After feature scaling the decision boundary in the right plot looks much better

Hard Margin vs Soft Margin SVM



- Hard margin: No data points are allowed in the margin areas
- If we strictly impose that all instances must be off the street and on the right side, this is called ***hard margin classification***
- There are two main issues with hard margin classification.
 - it only works if the data is linearly separable.
 - it is sensitive to outliers.

Hard Margin vs Soft Margin SVM

- When the data is **linearly separable**, and we don't want to have any **misclassifications**, we use SVM with a hard margin.
- However, when a linear boundary is not feasible, or we want to allow some misclassifications in the hope of achieving better generality, we can opt for a soft margin for our classifier
- This is called ***soft margin classification***
- The objective is to find a good balance between keeping the **street as large as possible** and **limiting the *margin violations***
- In soft margin, some instances can end up in the middle of the street or even on the wrong side



Hard Margin Classifier Objective

- SVM Optimization problem:
- The linear SVM classifier model predicts the class of a new instance $\mathbf{x}$ **by simply computing** the decision function $\mathbf{w}^T \mathbf{x} + b = w_1 x_1 + \dots + w_n x_n + b$.
- Form of equation defining the decision surface separating the classes is a hyperplane of the form:

$$\mathbf{w}^T \mathbf{x} + b = 0$$

Equation of a line: standard/matrix form

- $\mathbf{w}$ is a weight vector
- $\mathbf{x}$ is input vector
- b is bias
- Allows us to write
$$\mathbf{w}^T \mathbf{x} + b \geq 0 \text{ for } d_i = +1$$
$$\mathbf{w}^T \mathbf{x} + b < 0 \text{ for } d_i = -1$$

- w is the weight vector=slope/gradiant/coefficients= θ_0
- b is the scalar bias= intercept/ bias/constant= θ_1

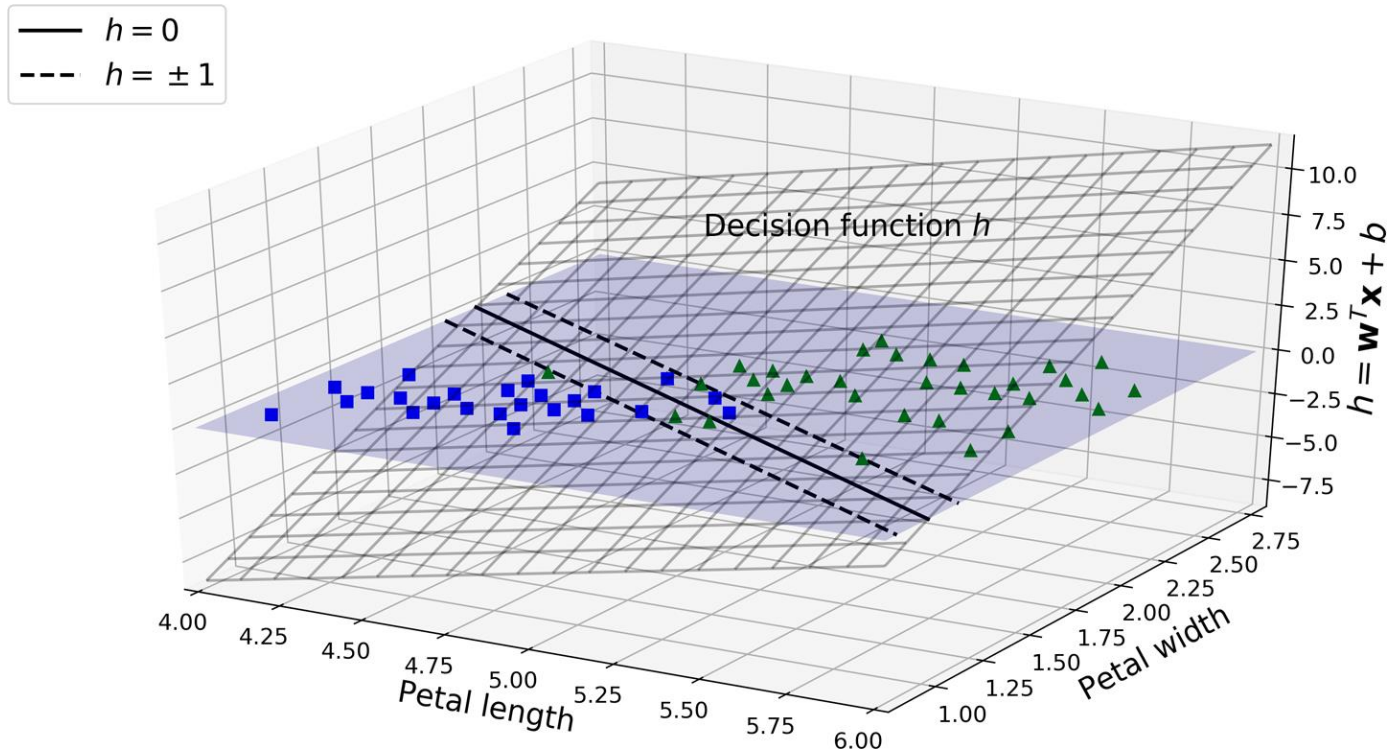


Hard Margin Classifier Objective

- If the result is positive, the predicted class $\hat{y}$ is *the positive class (1)*, and otherwise it is *the negative class (0)*;
- $y = 0$ if $\mathbf{w}^T \mathbf{x} + b < 0$,
1 if $\mathbf{w}^T \mathbf{x} + b \geq 0$



Hard Margin Classifier Objective



- it is a 2D plane because this dataset has two features (petal width and petal length).
- The decision boundary is the set of points where the decision function is equal to 0: it is the intersection of two planes, which is a straight line (represented by the thick solid line)

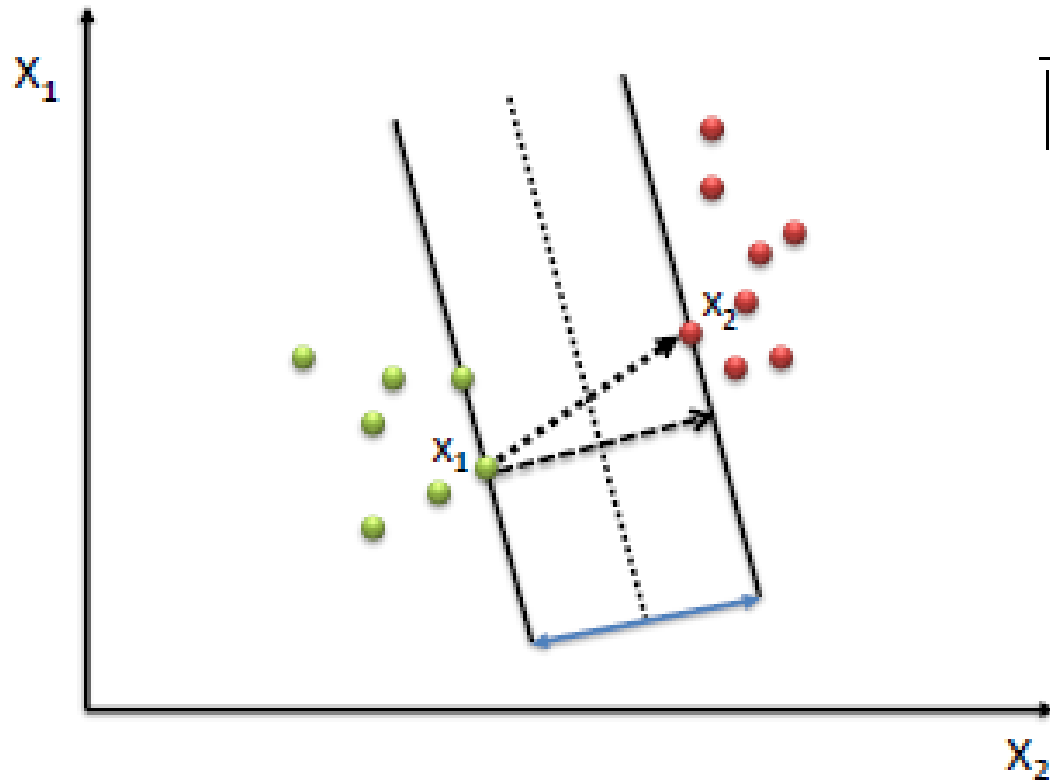


Hard Margin Classifier Objective

- The dashed lines represent the points where the decision function is equal to 1 or -1 :
- they are parallel and at equal distance to the decision boundary, and they form a margin around it.
- Training a linear SVM classifier means finding the values of **w and b** that make this margin **as wide as possible** while avoiding margin violations (hard margin) or limiting them (soft margin).



Hard Margin Classifier Objective



$$\frac{w}{\|w\|} \cdot (x_2 - x_1) = \text{width} = \frac{2}{\|w\|}$$

$$w \cdot x_2 + b = 1$$

$$w \cdot x_1 + b = -1$$

$$w \cdot x_2 + b - w \cdot x_1 - b = 1 - (-1)$$

$$w \cdot x_2 - w \cdot x_1 = 2$$

$$\frac{w}{\|w\|} (x_2 - x_1) = \frac{2}{\|w\|}$$

Goal: maximize

Hard Margin Classifier Objective

- Constrained Optimization problem:

$$\begin{aligned} & \text{maximize } \frac{2}{\|w\|} = \frac{2}{w^T w} \\ & \text{minimize } \frac{1}{2} w^T w \\ & \text{w, b} \end{aligned}$$

- Subject to the following constraint:

$$t^{(i)}(w^T x^{(i)} + b) \geq 1 \quad \text{for } i = 1, 2, \dots, m$$

- $t^{(i)}$ is -1 for negative instances, and $+1$ for positive instances
- w is the weight vector = θ_1 = slope/coefficient/gradient



Hard Margin Classifier Objective

- We are minimizing $\frac{1}{2} w^T w$, which is equal to $\frac{1}{2} \|w\|^2$, rather than minimizing $\|w\|$.
- Indeed, $\|w\|^2$ has a nice, simple derivative (it is just w), while $\|w\|$ is not differentiable at $w = 0$.



Hard Margin Classifier Objective

- Slope of the decision function is equal to the norm of the weight vector $\|\mathbf{w}\|$
- Smaller weight vector $\mathbf{w}$ means larger margin
- Minimize $\|\mathbf{w}\|$ to get a large margin, while avoiding margin violations

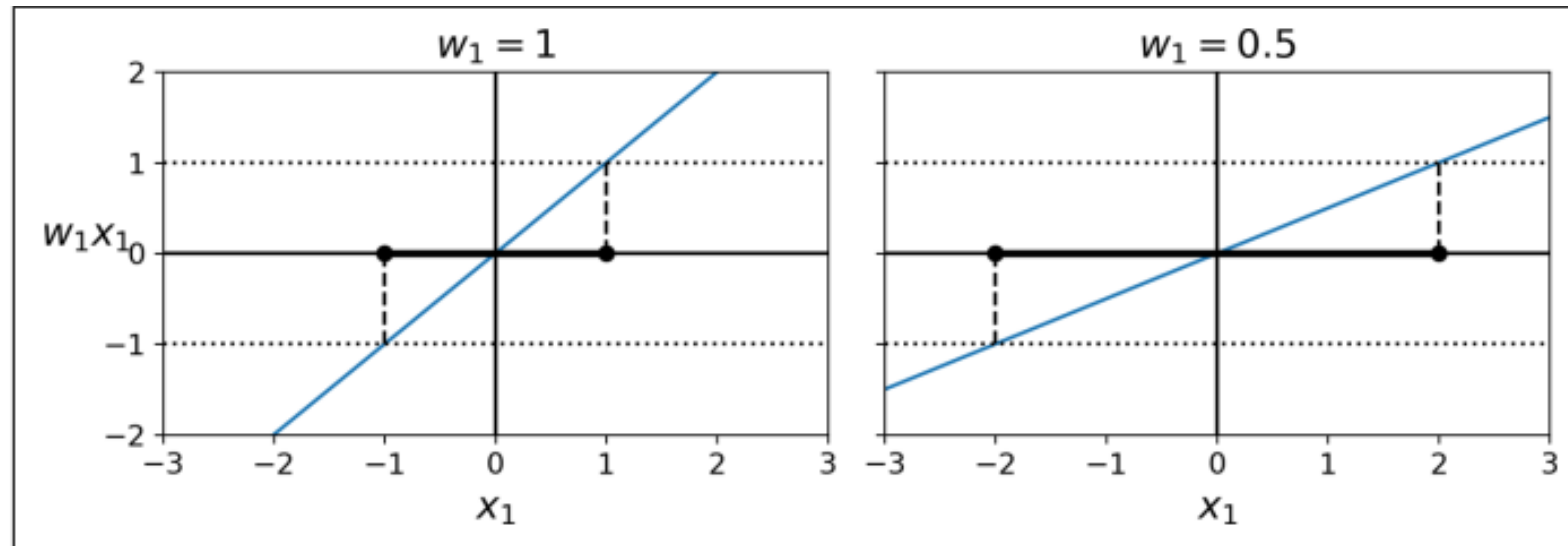
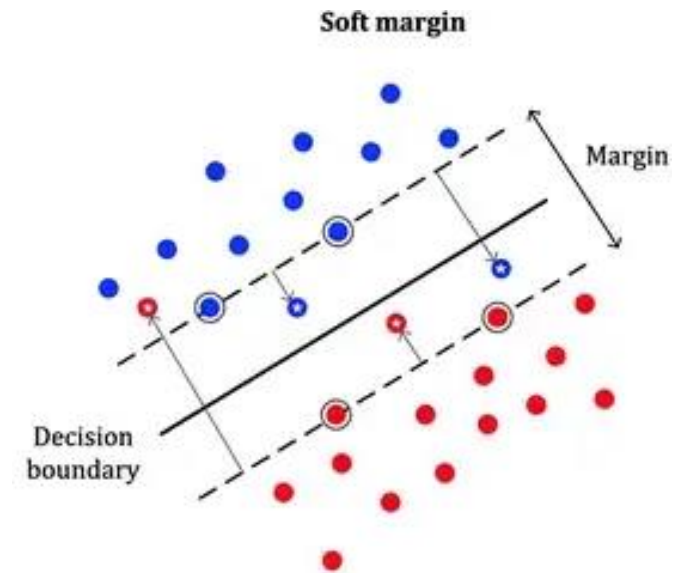


Figure 5-13. A smaller weight vector results in a larger margin

Soft Margin Classification

- Can't strictly enforce all training data to be on the correct side of the decision boundary (**hard margin classification**) – only works if it's **linearly separable**
- Adversely affected by outliers
- Keep margin as wide as possible, while limiting margin violations
- $C \rightarrow$ hyperparameter controlling **number of margin violations/ allowed misclassification/error**
- Penalty for allowed misclassification/error
- Penalty depends on the **distance of the misclassification point from its correct margin** = slack variable (ζ) zeta



Soft Margin Classifier Objective

- Introduce a slack variable $\zeta^{(i)} \geq 0$ for each instance – measure of how much the i^{th} instance is allowed to violate the margin: Hinge Loss

$$\underset{\mathbf{w}, b, \zeta}{\text{minimize}} \quad \frac{1}{2} \mathbf{w}^T \mathbf{w} + C \sum_{i=1}^m \zeta^{(i)}$$

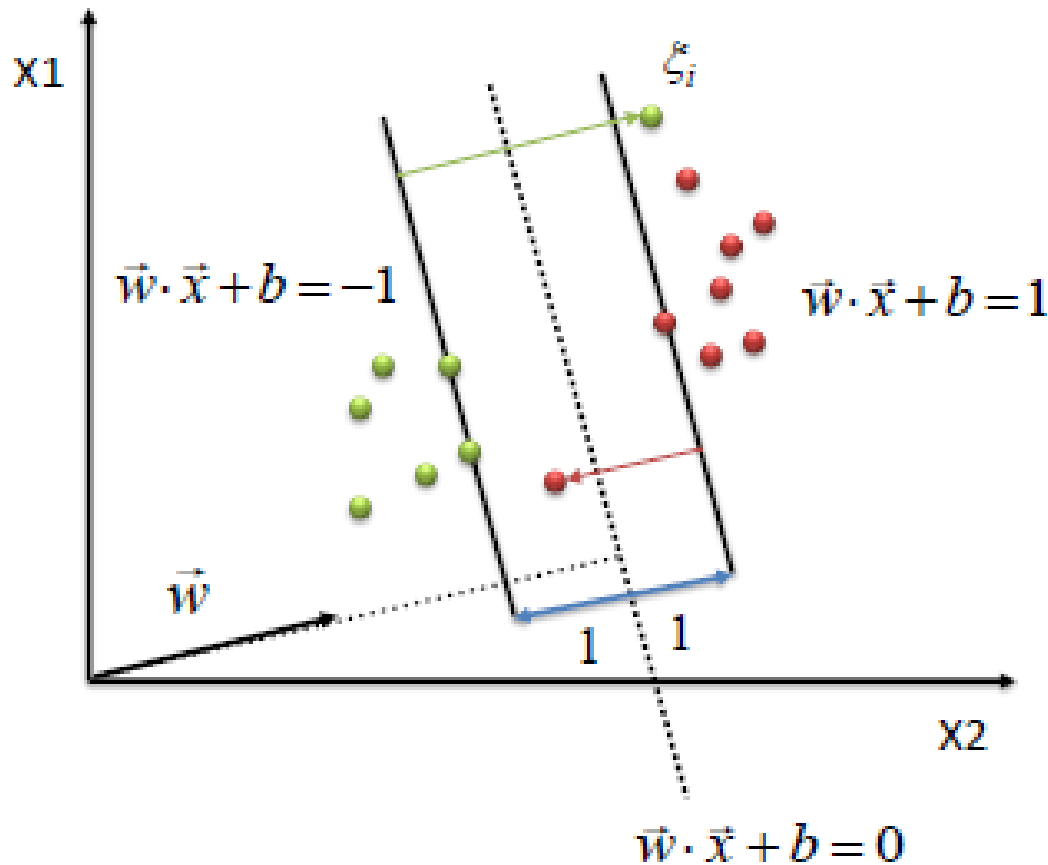
- Subject to the following constraint:

$$t^{(i)}(\mathbf{w}^T \mathbf{x}^{(i)} + b) \geq 1 - \zeta^{(i)} \quad \text{and} \quad \zeta^{(i)} \geq 0 \quad \text{for } i = 1, 2, \dots, m$$

- C = Regularization parameter = hyperparameter.



Soft Margin Classifier Objective



Constraint becomes :

$$y_i(w \cdot x_i + b) \geq 1 - \xi_i, \quad \forall x_i$$

$$\xi_i \geq 0$$

Objective function
penalizes for misclassified instances and those within the margin

$$\min \frac{1}{2} \|w\|^2 + C \sum_i \xi_i$$

C trades-off margin width and misclassifications

Soft Margin Classifier Objective

- To get the soft margin objective, we need to introduce a *slack variable* $\zeta(i) \geq 0$ for each instance:
 - $\zeta(i)$ measures how much the i -th instance is allowed to violate the margin.
- We now have two **conflicting objectives**:
 - make the slack variables as small as possible to reduce the margin violations
 - make $\frac{1}{2} w^T w$ as small as possible to increase the margin.
- This is where the C hyperparameter comes in: it allows us to define the tradeoff between these two objectives.

